

Heat

1. What is heat?

- A) Form of matter
- B) Form of energy
- C) Type of force
- D) Type of motion

Answer: B

2. Heat flows from

- A) Cold body to hot body
- B) Hot body to cold body
- C) Equal temperature bodies
- D) Vacuum to matter

Answer: B

3. Temperature is the measure of

- A) Heat energy
- B) Degree of hotness or coldness
- C) Mass of body
- D) Size of body

Answer: B

4. SI unit of heat is

- A) Joule
- B) Watt
- C) Kelvin
- D) Newton

Answer: A

5. The SI unit of temperature is

- A) Joule
- B) Kelvin
- C) Watt
- D) Pascal

Answer: B

6. Heat transfer in solids mainly occurs by

- A) Conduction
- B) Convection
- C) Radiation
- D) Reflection

Answer: A

7. Heat transfer in liquids and gases mainly occurs by

- A) Conduction
- B) Convection
- C) Radiation
- D) Refraction

Answer: B

8. Heat transfer that does not require a medium is

- A) Conduction
- B) Convection
- C) Radiation
- D) Diffusion

Answer: C

9. Heat from the Sun reaches the Earth by

- A) Conduction
- B) Convection
- C) Radiation
- D) Reflection

Answer: C

10. Metals are generally

- A) Bad conductors
- B) Good conductors
- C) Insulators
- D) Non-metals

Answer: B

11. Which is a good conductor of heat?

- A) Wood

B) Copper

C) Plastic

D) Rubber

Answer: B

12. Which is a poor conductor of heat?

A) Aluminium

B) Copper

C) Iron

D) Wood

Answer: D

13. Woollen clothes keep us warm because

A) Wool produces heat

B) Wool traps air

C) Wool reflects heat

D) Wool absorbs heat

Answer: B

14. Air is a

A) Good conductor

B) Poor conductor

C) Metal

D) Liquid

Answer: B

15. Convection currents occur in

A) Solids

B) Liquids and gases

C) Vacuum

D) Metals only

Answer: B

16. Convection cannot occur in

A) Liquids

B) Gases

C) Solids

D) Water

Answer: C

17. Sea breeze occurs during

A) Night

B) Day

C) Winter

D) Rain

Answer: B

18. Land breeze occurs during

A) Day

B) Night

C) Noon

D) Summer

Answer: B

19. White clothes are preferred in summer because they

A) Absorb heat

B) Reflect heat

C) Produce heat

D) Store heat

Answer: B

20. Black surfaces are

A) Good reflectors

B) Good absorbers

C) Poor absorbers

D) Poor emitters

Answer: B

21. The process of heat transfer without movement of molecules is

A) Conduction

B) Convection

C) Radiation

D) Diffusion

Answer: A

22. The process of heat transfer by movement of molecules is

- A) Conduction
- B) Convection
- C) Radiation
- D) Absorption

Answer: B

23. Thermal equilibrium occurs when

- A) Heat stops flowing
- B) Temperature becomes equal
- C) Heat increases
- D) Heat decreases

Answer: B

24. Specific heat capacity is the heat required to raise temperature of

- A) 1 g by 1°C
- B) 1 kg by 1°C
- C) 10 kg by 1°C
- D) 1 kg by 10°C

Answer: B

25. Unit of specific heat capacity is

- A) J/kg K
- B) J/K
- C) W/m
- D) N/kg

Answer: A

26. Heat capacity is the heat required to raise temperature of

- A) Unit mass
- B) Whole body
- C) One gram
- D) One litre

Answer: B

27. SI unit of heat capacity is

- A) J/K

B) J/kg

C) K/J

D) N/K

Answer: A

28. Latent heat is heat required for

A) Temperature rise

B) State change

C) Pressure change

D) Volume change

Answer: B

29. During melting, temperature

A) Increases

B) Decreases

C) Remains constant

D) Fluctuates

Answer: C

30. Melting point of ice is

A) 0°C

B) 10°C

C) 50°C

D) 100°C

Answer: A

31. Boiling point of water is

A) 0°C

B) 50°C

C) 100°C

D) 200°C

Answer: C

32. Conversion of solid to liquid is

A) Freezing

B) Melting

C) Condensation

D) Sublimation

Answer: B

33. Conversion of liquid to solid is

A) Freezing

B) Boiling

C) Melting

D) Condensation

Answer: A

34. Conversion of liquid to gas is

A) Freezing

B) Boiling

C) Melting

D) Deposition

Answer: B

35. Conversion of gas to liquid is

A) Condensation

B) Freezing

C) Melting

D) Sublimation

Answer: A

36. Conversion of solid directly to gas is

A) Condensation

B) Sublimation

C) Freezing

D) Boiling

Answer: B

37. Specific latent heat formula is

A) $L = Q/m$

B) $L = m/Q$

C) $L = Q \times m$

D) $L = Q + m$

Answer: A

38. Unit of latent heat is

- A) J/kg
- B) J
- C) K
- D) W

Answer: A

39. Steam burns are more severe because of

- A) Higher temperature
- B) Latent heat
- C) Conduction
- D) Radiation

Answer: B

40. Aluminium is used for utensils because it

- A) Is expensive
- B) Is good conductor
- C) Is insulator
- D) Is heavy

Answer: B

41. Mercury is used in thermometers because

- A) Good conductor
- B) Poor conductor
- C) Solid
- D) Colourless

Answer: A

42. Earthen pots keep water cool due to

- A) Conduction
- B) Evaporation
- C) Radiation
- D) Condensation

Answer: B

43. Expansion occurs due to

- A) Loss of heat

B) Gain of heat

C) Pressure

D) Gravity

Answer: B

44. Expansion is maximum in

A) Solids

B) Liquids

C) Gases

D) Metals

Answer: C

45. Expansion is least in

A) Solids

B) Liquids

C) Gases

D) Air

Answer: A

46. Railway tracks have gaps to allow

A) Cooling

B) Expansion

C) Contraction

D) Painting

Answer: B

47. Chimneys are tall because

A) Decoration

B) Convection

C) Conduction

D) Radiation

Answer: B

48. Air rises when heated because it becomes

A) Heavy

B) Lighter

C) Colder

D) Denser

Answer: B

49. Sea breeze is caused due to

A) Conduction

B) Convection

C) Radiation

D) Reflection

Answer: B

50. Land breeze occurs because

A) Land cools faster

B) Sea cools faster

C) Air disappears

D) Heat stops

Answer: A

51. Heat energy increases

A) Potential energy only

B) Kinetic energy only

C) Both energies

D) None

Answer: B

52. Heating a substance causes

A) Temperature rise

B) Pressure loss

C) Volume loss

D) Colour change

Answer: A

53. Cooling a substance causes

A) Temperature fall

B) Temperature rise

C) Heat gain

D) Mass gain

Answer: A

54. Thermal equilibrium means

- A) Equal heat
- B) Equal temperature
- C) Equal mass
- D) Equal size

Answer: B

55. Ice absorbs heat during

- A) Freezing
- B) Melting
- C) Condensation
- D) Deposition

Answer: B

56. Water releases heat during

- A) Boiling
- B) Freezing
- C) Melting
- D) Heating

Answer: B

57. Heat transfer by direct contact is

- A) Conduction
- B) Convection
- C) Radiation
- D) Evaporation

Answer: A

58. Radiation can occur in

- A) Vacuum
- B) Liquid
- C) Solid
- D) Gas

Answer: A

59. Dark coloured surfaces

- A) Reflect heat

- B) Absorb heat
- C) Produce heat
- D) Destroy heat

Answer: B

60. Light coloured surfaces

- A) Absorb heat
- B) Reflect heat
- C) Destroy heat
- D) Produce heat

Answer: B

61. The bottom of cooking utensils is blackened to

- A) Reflect heat
- B) Absorb heat
- C) Reduce heat
- D) Prevent heat

Answer: B

62. Absolute zero temperature is

- A) 0°C
- B) -273°C
- C) 100°C
- D) 273°C

Answer: B

63. Absolute zero is the temperature where

- A) Volume increases
- B) Pressure increases
- C) Gas volume theoretically zero
- D) Gas melts

Answer: C

64. Heat energy is measured using

- A) Thermometer
- B) Calorimeter
- C) Barometer

D) Ammeter

Answer: B

65. Temperature is measured using

A) Thermometer

B) Ammeter

C) Voltmeter

D) Hygrometer

Answer: A

66. Conduction is common in

A) Solids

B) Liquids

C) Gases

D) Vacuum

Answer: A

67. Convection is common in

A) Solids

B) Liquids

C) Vacuum

D) Metals

Answer: B

68. Radiation travels in the form of

A) Waves

B) Molecules

C) Atoms

D) Particles

Answer: A

69. Heat energy causes molecules to

A) Slow down

B) Stop moving

C) Move faster

D) Freeze

Answer: C

70. Melting of ice requires

- A) Loss of heat
- B) Gain of heat
- C) No heat
- D) Pressure only

Answer: B

71. Freezing releases

- A) Heat
- B) Light
- C) Sound
- D) Pressure

Answer: A

72. Vaporization requires

- A) Heat absorption
- B) Heat release
- C) No heat
- D) Cooling

Answer: A

73. Condensation releases

- A) Heat
- B) Light
- C) Pressure
- D) Energy loss

Answer: A

74. During boiling temperature

- A) Increases
- B) Decreases
- C) Remains constant
- D) Doubles

Answer: C

75. During melting temperature

- A) Constant

- B) Rising
- C) Falling
- D) Random

Answer: A

76. Heat transfer through empty space is

- A) Conduction
- B) Convection
- C) Radiation
- D) Evaporation

Answer: C

77. A good insulator of heat is

- A) Copper
- B) Aluminium
- C) Plastic
- D) Iron

Answer: C

78. Wood is used for handles of utensils because it

- A) Conducts heat
- B) Insulates heat
- C) Reflects heat
- D) Emits heat

Answer: B

79. The movement of warm fluid upward is due to

- A) High density
- B) Low density
- C) No density
- D) Pressure

Answer: B

80. Cold air is

- A) Denser
- B) Lighter
- C) Hotter

D) Empty

Answer: A

81. Hot air balloon rises due to

A) Conduction

B) Convection

C) Radiation

D) Expansion

Answer: B

82. The purple streak experiment with KMnO_4 shows

A) Conduction

B) Convection

C) Radiation

D) Sublimation

Answer: B

83. Heat transfer in metals mainly occurs due to

A) Free electrons

B) Molecules only

C) Atoms only

D) Light

Answer: A

84. The SI unit of latent heat is

A) J/kg

B) J

C) K

D) W

Answer: A

85. Heat capacity depends on

A) Mass

B) Colour

C) Shape

D) Height

Answer: A

86. Specific heat capacity depends on

- A) Substance nature
- B) Mass
- C) Volume
- D) Colour

Answer: A

87. Heat is transferred until

- A) Pressure equalizes
- B) Temperature equalizes
- C) Mass equalizes
- D) Volume equalizes

Answer: B

88. When heat is added molecules

- A) Vibrate faster
- B) Stop moving
- C) Disappear
- D) Become solid

Answer: A

89. When heat is removed molecules

- A) Slow down
- B) Speed up
- C) Disappear
- D) Melt

Answer: A

90. Snow acts as

- A) Conductor
- B) Insulator
- C) Radiator
- D) Heater

Answer: B

91. Igloos are warm inside because snow is

- A) Conductor

B) Insulator

C) Radiator

D) Heater

Answer: B

92. Air trapped in wool acts as

A) Conductor

B) Insulator

C) Heater

D) Cooler

Answer: B

93. The flow of heat depends on

A) Temperature difference

B) Mass difference

C) Volume difference

D) Colour difference

Answer: A

94. The study of heat is called

A) Thermodynamics

B) Mechanics

C) Optics

D) Magnetism

Answer: A

95. A polished surface

A) Absorbs heat

B) Reflects heat

C) Produces heat

D) Stores heat

Answer: B

96. Radiation travels with

A) Medium

B) No medium

C) Water

D) Metal

Answer: B

97. Latent heat does not change

A) State

B) Temperature

C) Mass

D) Pressure

Answer: B

98. Water at 100°C changing to steam needs

A) Latent heat of fusion

B) Latent heat of vaporization

C) Specific heat

D) Conduction

Answer: B

99. Ice changing to water needs

A) Latent heat of fusion

B) Latent heat of vaporization

C) Conduction

D) Radiation

Answer: A

100. Heat is transferred until bodies reach

A) Thermal equilibrium

B) High pressure

C) Low pressure

D) Motion